

Aim

To find the Continuous Time Fourier Transform (CTFT) of a rectangular pulse and the cosine function

Theory

The Continuous Time Fourier Transform (CTFT) for a given function is given by:

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Where $f(t)$ is the function, j is the imaginary unit, ω is the angular frequency and t is the time period.

The CTFT is the signal analysis tool that can be used to represent time domain function in the frequency domain and represent them as algebraic functions. It helps us break apart a signal and see what a function looks like at each frequency.

MATLAB Code

```
close all;
clear vars;

t = -50: 10^-2: 50;
w = -10 * pi: 10^-2 :10 * pi;

f1_ = (t >= -5) & (t <= 5);
f2_ = cos (10 * t);

F1_ = zeros(1, length(w));

for k = 1: length(w)
    F1_(k) = trapz (t, f1_ .* exp(-1 * (sqrt(-1) * w(k) * t)));
end

F2_ = zeros(1, length(w));

for k = 1: length(w)
    F2_(k) = trapz (t, f2_ .* exp(-1 * (sqrt(-1) * w(k) * t)));
end

figure;

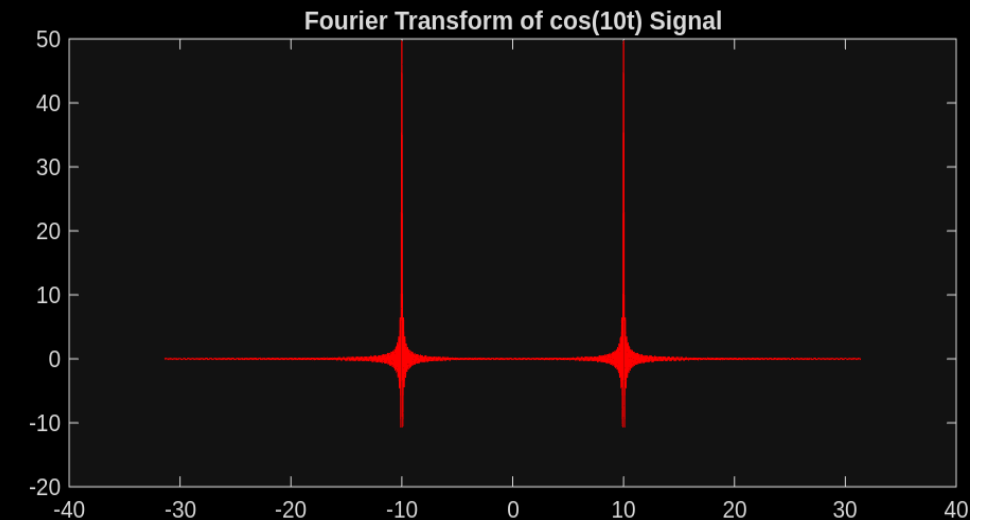
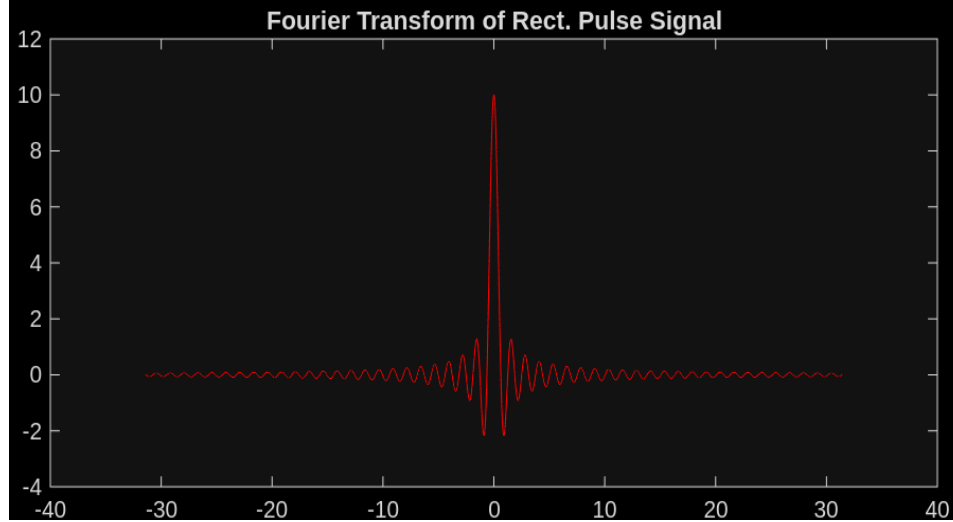
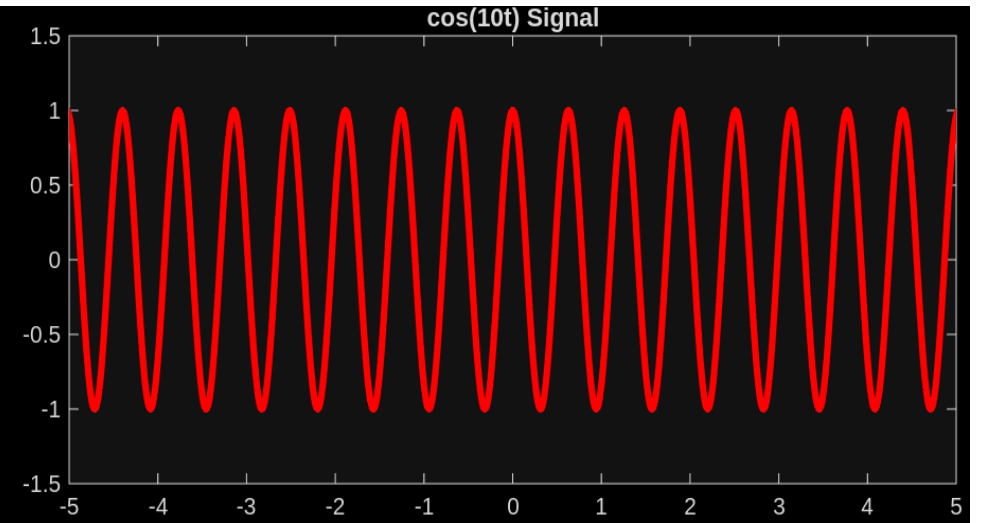
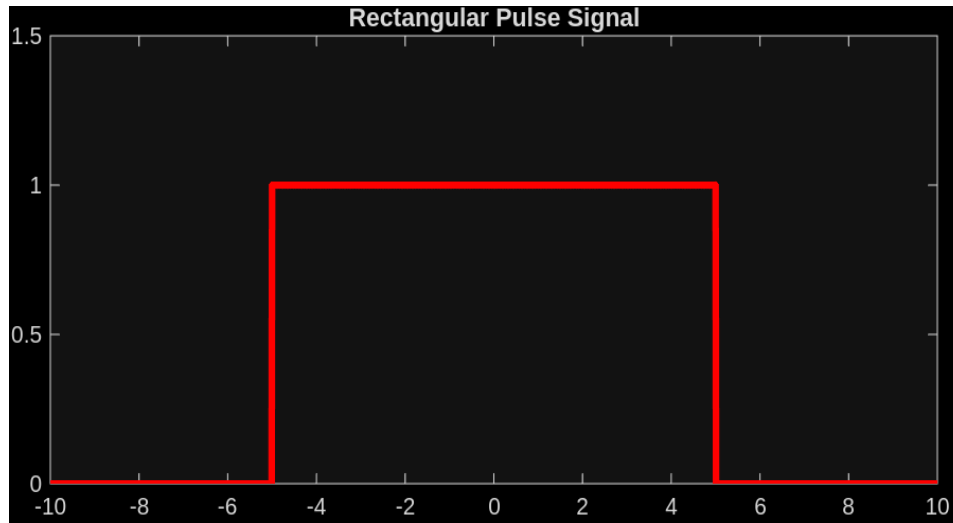
subplot(2,2,1)
plot(t, f1_,'r', LineWidth=3)
title('Rectangular Pulse Signal')
xlim([-10, 10])
ylim([0, 1.5])

subplot(2,2,2)
plot(t, f2_, 'r', LineWidth=3)
title("cos(10t) Signal")
xlim([-5, 5])
ylim([-1.5, 1.5])

subplot(2,2,3)
plot(w, F1_, "r")
title('Fourier Transform of Rect. Pulse Signal')

subplot(2,2,4)
plot(w, F2_, 'r')
title('Fourier Transform of cos(10t) Signal')
```

Results



Conclusions

In this experiment, we have successfully converted two functions from the time domain to the frequency domain and have plotted them against their angular frequency. We see that the CTFT breaks down a time-domain signal into its constituent frequencies and their magnitude and phase spectra.